PROJECT

7.2

CONNECT A GITHUB REPO WITH AWS

DEVOPS SERIES

Prepared by:
Amrize JK
Certified Solutions Architect Associate

Objective

This project aims to connect a cloud-hosted web application with a GitHub repository for version control and remote collaboration.

It involves setting up Git on an EC2 instance, securely pushing code changes to GitHub using personal access tokens, and managing code updates through Git commits.

Services

Amazon EC2 - Hosts the web application and development environment in the cloud.

AWS IAM - Provides secure user access to the EC2 instance.

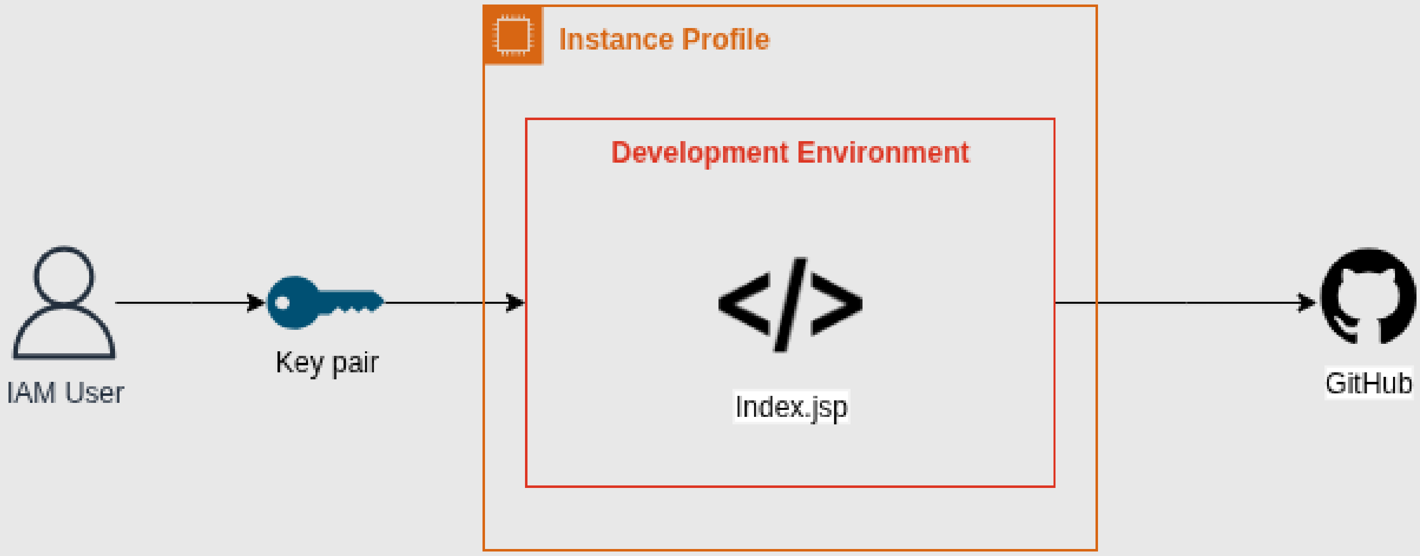
Git - Handles version control and tracks code changes locally.

GitHub - Stores the project repository and manages code collaboration remotely.

GitHub PAT - Authenticates secure Git operations between EC2 and GitHub.

VSCode - Used as the code editor and interface for Git operations.

Architecture Diagram



An IAM user uses a key pair to access the EC2 instance, which has an instance profile for AWS permissions. Inside, the development environment hosts the app (e.g., Index.jsp). Code is edited via VSCode and pushed to GitHub for version control and collaboration.

Steps to Implement

1. Set up your web app in the cloud:
Launch an EC2 instance, connect using VSCode via Remote-SSH, and ensure your app environment is ready.

2. Install Git on EC2:
Use terminal commands to install Git (sudo yum install git or sudo apt install git) and verify with git --version.

3. Set up GitHub connection:
Create a new repo on GitHub, configure your EC2 with git config for username and email, and initialize a local Git repo with git init.

4. Commit and push code to GitHub:
Add your files with git add ., commit using git commit -m "initial commit", and push using your GitHub repo's HTTPS URL.

5. Generate and use a GitHub PAT:
Create a Personal Access Token from GitHub, and use it as your password during the git push operation for secure authentication.

6. Make your second commit:
Modify any file, add, commit, and push again to understand the workflow better.

7. Create a README file:
Add a README.md to document your project, commit it, and push the changes to reflect them on GitHub.

Story Time

IAM (The Gatekeeper of Permissions):
IAM is your mighty gatekeeper who hands out magical badges (permissions) to chosen warriors (users). Only those with the right badge can enter the EC2 realm and command cloud resources.

EC2 (The Cloud Coding Castle):
EC2 is your virtual fortress in the sky where your code lives and breathes. It's the home base where you build, run, and test your web app like a true digital wizard.

Git (The Time-Travel Scroll):
Git is your enchanted scroll that remembers every spell you ever cast. With it, you can rewind, branch, or replay your code adventures anytime.

GitHub (The Spellbook Vault):
GitHub is your secure vault in the cloud where you store and share your spellbooks (repositories). It protects your code and lets others collaborate on your magical creations.

GitHub PAT (The Secret Cloud Key):
The Personal Access Token is your secret cloud key. Instead of shouting passwords at the gate, you quietly use this enchanted token to prove your identity and push your spells to GitHub.

VSCode (The Wizard's Command Console):
VSCode is your personal magic console, where you write spells (code), run commands, and control the entire flow of your development journey with just a flick of your wand.

BY

AMRIZE J.K

THANK YOU!

DEVOPS ENGINEER

IN THE MAKING